PALTHEPU SAI SAHITH

+91 7702779132 | palthepusaisahith66@gmail.com | LinkedIn | GitHub

SUMMARY

Software engineer focused on backend and reliable systems, with experience building stateful AI platforms, data-processing pipelines, and ML applications. Strong in Python, C++, Java, PostgreSQL, FastAPI, Docker, and React, with research experience in computational topology and scientific visualization.

EDUCATION

Manipal Institute of Technology, Bengaluru 2025
B.Tech in Computer Science – Artificial Intelligence & Machine Learning

SKILLS

Languages	Python, C++, Java, SQL, TypeScript/JavaScript
Backend & Systems	FastAPI, PostgreSQL, REST APIs, Docker, Git, OpenTelemetry
AI/ML	TensorFlow, PyTorch, OpenCV, Computer Vision
Research	Computational Topology, Scientific Visualization, ParaView/TTK

EXPERIENCE

International Institute of Information Technology, Hyderabad Jan 2026 – Present
Research Assistant, CVIT

- Conduct research on computational-topology methods for comparing scientific scalar-field data using persistence distortion distance (PDD), extremum graphs, merge trees, and topology-based analysis.
- Extend C++ PDD implementations and build Python automation for pairwise experiments, result aggregation, plotting, and numerical validation across scientific time steps.
- Automate ParaView/TTK persistence simplification, Morse–Smale analysis, topology edits, and graph exports; validate thresholds and intermediate outputs before large batch experiments.

COE Software Technologies Pvt Ltd, Hyderabad Aug 2023 – Oct 2023
Software Engineering Intern

- Built backend modules for a student data-management system, improving retrieval speed by 30% while reliably handling 2,000+ records.
- Automated data storage, updates, validation, and security workflows, improving operational accuracy and scalability for institutional users.
- Collaborated with cross-functional teams to integrate ML workflows and support deployment and real-time accessibility across schools and colleges.

PROJECTS

ChaosAgent – Reliability Evaluation Platform for Tool-Using AI Agents GitHub

- Built a stateful reliability-evaluation platform that injects deterministic failures into synthetic business workflows and evaluates agent recovery using authoritative state and event evidence.
- Engineered PostgreSQL-backed orchestration with safe leasing, idempotent mutations, and durable approvals to prevent duplicate effects during retries and ambiguous failures.
- Exposed experiments through FastAPI REST and replay-safe SSE with a React causal dashboard; added deterministic evaluators, OpenTelemetry, abuse tests, and an 882-test Python suite.

PraxisMirror – Evidence-Grounded Object-Centric Process Analytics *Python, OCEL 2.0, Process Mining*

- Built a deterministic OCEL 2.0 ingestion and analytics system preserving typed attributes, temporal history, qualified object-event relationships, source lineage, and reproducible snapshots.
- Designed evidence contracts and semantic compatibility checks to prevent metrics from being reused across incompatible units, cohorts, perspectives, or computation revisions.
- Implemented object-centric analytics for container trajectories, elapsed-time populations, missing departures, and rescheduling cohorts, backed by 90 automated tests covering deterministic computation, evidence validation, and adversarial cases.

PneumoNet – Federated Pneumonia Detection *TensorFlow Federated, VGG16, Streamlit*

- Built a federated-learning pipeline for pneumonia detection from decentralized chest X-rays using VGG16 transfer learning without centralizing raw image datasets.
- Implemented preprocessing, distributed training, evaluation, and Streamlit inference workflows to deliver an end-to-end ML application.